# Solution 1 Classifying Lie algebras of dim at most 3

**Problem 1.1** — Classify $\mathbb{C}$ Lie algebras of dim at most 3, up to isomorphism.

Observe that "up to isomorphism" just means "up to basis change."

**Case** (Dimension 0) —

There is only one vector space of dimension 0, namely $\{0\}$.

**Case** (Dimension 1) —

There is only one Lie algebra of dimension 1, $\mathbb{C}$, and all Lie brakets must be 0 by antisymmetry.

**Case** (Dimension 2) —

Suppose $L = \operatorname{span}(x, y)$; then $[L, L] = \operatorname{span}([x, y])$ has one dimension; thus let $0 \neq x \in [L, L]$, and pick any $y$ to fill the basis. Then, a Lie algebra is completely specified by the value of $\alpha \in \mathbb{C}$ of $[x, y] = \alpha x$. There are then two cases: if $\alpha \neq 0$, one can scale $y' = y/\alpha$ to make $\alpha = 1$. Otherwise, $[x, y] = 0$, and $L$ is abelian.

There are two solutions:

1. $L$ is abelian.
2. $L$ is spanned by $x$ and $y$ with $[x, y] = x$.

**Case** (Dimension 3, Abelian) —

There is only one abelian Lie algebra of dimension 3.

**Case** (Dimension 3, Commutator has dimension 1) —

In this case, let $x \in [L, L]$; then the Lie algebra is completely specified by the values of $\alpha, \beta, \gamma \in \mathbb{C}$ of

$$[x, y] = \alpha x, \quad [x, z] = \beta x, \quad [y, z] = \gamma x.$$

I claim I can force either $\alpha = 0$ or $\beta = 0$ through a basis change. If both are currently nonzero, then set $y' = y/\alpha - z/\beta$; then

$$[x, y'] = [x, y]/\alpha - [x, z]/\beta = 0$$

Okay, so WLOG $\beta = 0$ above. Next, we have four cases:

- If $\alpha = \gamma = 0$, then $[L, L] = 0$, contradiction.
- If $\alpha \neq 0$ but $\gamma = 0$, then let $y' = y/\alpha$ such that the Lie algebra is

$$[x, y] = x, \quad [x, z] = 0, \quad [y, z] = 0.$$

- If $\alpha = 0$ but $\gamma \neq 0$, then let $z' = z/\gamma$ such that the Lie algebra is

$$[x, y] = 0, \quad [x, z] = 0, \quad [y, z] = x.$$

  This is distinct from the previous Lie algebra, because $\mathrm{ad}_x = 0$ in this algebra whereas $\mathrm{ad}_x \neq 0$ in the previous algebra.

- If $\alpha \neq 0$ and $\gamma \neq 0$, then let $y' = y/\alpha$ and $z' = z\alpha/\gamma$ such that the Lie algebra is

$$[x, y] = x, \quad [x, z] = 0, \quad [y, z] = x.$$

  Then let $z' = z + x$ such that the Lie algebra is

$$[x, y] = x, \quad [x, z] = 0, \quad [y, z] = 0.$$

  This is clearly the same as the first case.

So there are two distinct solutions in this case.

**Case** (Dimension 3, Commutator has dimension 2, the restriction to the commutator is abelian) —

To write out the case that we're in, let $x, y \in [L, L]$ be linearly independent and $z$ be linearly independent from $x, y$; we're in the case where

$$[x, y] = 0, \qquad \mathrm{ad}_z(L) = [L, L]$$

Well okay, I can perform arbitrary linear transformations on $x, y, z$. Linear transformations on $x$ and $y$ will conjugate $\mathrm{ad}_z(L)$, adding copies of $x, y$ to $z$ will do nothing to $\mathrm{ad}_z$, and scaling $z$ will scale $\mathrm{ad}_z$.

By Jordan canonical form, I can write $\mathrm{ad}_z$ in one of two ways:

- We have

$$\mathrm{ad}_z = \begin{pmatrix} \alpha & 0 \\ 0 & \beta \end{pmatrix}$$

  (We're not writing the $z$-dimension of the domain or range because we know $[z, z] = 0$ and $z \notin [L, L]$). Note neither $\alpha$ nor $\beta$ can be zero, because $\mathrm{ad}_z(L) = [L, L]$ exactly. Then set $z' = z/\alpha$ to make $\mathrm{ad}_z$ of the form

$$\begin{pmatrix} 1 & 0 \\ 0 & \beta \end{pmatrix}, \qquad \beta \in \mathbb{C} \setminus \{0\}$$

- $\mathrm{ad}_z$ is a Jordan block of size 2. In this case, $\mathrm{ad}_z$ is of the form

$$\mathrm{ad}_z = \begin{pmatrix} \alpha & 1 \\ 0 & \alpha \end{pmatrix}$$

  Scale $z$ to make the generalized eigenvalue equal to 1; then after a basis change, $\mathrm{ad}_z$ is of the form

$$\mathrm{ad}_z = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$$

**Case** (Dimension 3, Commutator has dimension 2, the restriction to the commutator is not abelian) —

To write out the case that we're in, let $x, y \in [L, L]$ be linearly independent and $z$ be linearly independent from $x, y$; by the dimension 2 case, WLOG we're in the case where

$$[x, y] = x, \qquad \mathrm{ad}_z(L) \subset [L, L]$$

The trick now is to notice that this is impossible. Pick any $z$ to fill out the basis, such that we can talk about $x$, $y$, and $z$ components. We know $y \in [L, L]$, but clearly $x$ and $y$ cannot be combined to create a $y$. Thus either $[x, z]$ or $[y, z]$ must contain a $y$ component.

- If $[x, z]$ contains a $y$ component, notice that

$$\underbrace{[x, [y, z]] + [y, [z, x]]}_{\in \mathbb{C}x} + \underbrace{[z, [x, y]]}_{\notin \mathbb{C}x} = 0$$

  contradiction.

- If $[y, z]$ contains a $y$ component, notice that

$$\underbrace{[x, [y, z]] + [y, [z, x]]}_{\in \mathbb{C}x} + [z, [x, y]] = 0$$

  thus $[x, z] \in \mathbb{C}x$. Scale $z$ such that $[x, y] = [x, z] = x$. The Jacobi identity now reads

$$0 = [x, [y, z]] + \underbrace{[y, [z, x]]}_{x} + \underbrace{[z, [x, y]]}_{x} = [x, [y, z]]$$

  So $[y, z]$ actually cannot contain any $y$ components, contradiction.

**Case** (Dimension 3, Commutator has dimension 3) —

In this case, $[L, L] = L$, and all adjoint maps have rank 2. Pick any $x \in L$. If $x$ is nilpotent, then because it is rank 2 we can write

$$\mathrm{ad}_x = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}$$

in some basis $\{x, y, z\}$. But then we see $[x, y] = x$ for instance, so $\mathrm{ad}_y(x) = -x$. Either way, I can find an $x \in L$ such that $\mathrm{ad}_x$ has an eigenvector with nonzero eigenvalue. By scaling, that eigenvector, $y$, has eigenvalue 1.

Then, because we know $[L, L] = L$, $[x, x] = 0$. We know $x = [a, b]$ for some $a, b \in L$, so if we do some shenanigans:

$$\mathrm{ad}_{[a,b]} = [\mathrm{ad}_a, \mathrm{ad}_b]$$

Thus $\mathrm{Tr}(\mathrm{ad}_x) = 0$ and we know $\mathrm{ad}_x$ has a different eigenvector with eigenvalue $-1$ called $z$. So now we've got:

$$[x, y] = y, \qquad [z, x] = z.$$

Fine. Now, we apply the Jacobi identity to get

$$[x, [y, z]] + [y, [z, x]] + [z, [x, y]] = 0 \implies [x, [y, z]] = [y, z] + [z, y] = 0$$

Well, the only thing annihilated by $\mathrm{ad}_x$ is $x$ itself, so $[y, z] \in \mathbb{C}x$. This can't be zero, so we might as well scale both $y$ and $z$ again such that $[y, z] = x$. Then, we have

$$[x, y] = y, \qquad [z, x] = z, \qquad [y, z] = x.$$

**Problem 1.2** — Classify real Lie algebras of dim at most 3.

No modifications need to be made to the 1, 2, abelian 3-dimensional, and 3-dimensional with $\dim[L, L] = 1$ cases; in those cases I did not use anything involving the algebraic completeness of $\mathbb{C}$ or anything, and the proof remains valid over $\mathbb{R}$.

**Case** (Dimension 3, Commutator has dimension 2, the restriction to the commutator is abelian) —

To write out the case that we're in, let $x, y \in [L, L]$ be linearly independent and $z$ be linearly independent from $x, y$; we're in the case where

$$[x, y] = 0, \qquad \mathrm{ad}_z(L) = [L, L]$$

Well okay, I can perform arbitrary linear transformations on $x, y, z$. Linear transformations on $x$ and $y$ will conjugate $\mathrm{ad}_z(L)$, adding copies of $x, y$ to $z$ will do nothing to $\mathrm{ad}_z$, and scaling $z$ will scale $\mathrm{ad}_z$.

I can use Jordan canonical form, I just need to stay aware that not all (complex) basis changes are possible. Remember that $\mathrm{ad}_z(L)$ must have rank 2. There are two possibilities:

- $L$ is diagonalizable. There are two sub-cases:
  - Both eigenvalues are real; say $\alpha$ and $\beta$. WLOG $|\alpha| > |\beta|$. We can then scale such that $\alpha = 1$; in this case, $\beta \in [-1, 1]$. To re-iterate:

    $$[z, x] = x, \qquad [z, y] = \beta y$$

  - Both eigenvalues are complex, in which case they must be conjugate by the characteristic polynomial. Say $\alpha$ and $\overline{\alpha}$. Scale $z$ such that they are roots of unity, $e^{i\theta}$ and $e^{-i\theta}$. Then, by a change of basis, one can write this as

    $$\begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$$

    i.e.

    $$[z, x] = x\cos\theta - y\sin\theta, \qquad [z, y] = x\sin\theta + y\cos\theta$$

- $L$ is nilpotent with generalized eigenvalue $\alpha \in \mathbb{R}$ (which must be real by e.g. the characteristic polynomial). We scale in the same as before, yielding

  $$[z, x] = x, \qquad [z, y] = x + y$$

The above possibilities are exhaustive and disjoint, because $[L, L]$ is basis-invariant, and as we showed above any possible $\mathrm{ad}_z$ is some combination of a basis change and scaling.

**Case** (Dimension 3, Commutator has dimension 3) —

In this case, $[L, L] = L$, and all adjoint maps have rank 2. Following the $\mathbb{C}$ proof, I can find an $x \in L$ such that $\mathrm{ad}_x$ has an eigenvector with nonzero eigenvalue $\lambda$.

Then, because we know $[L, L] = L$, $[x, x] = 0$. We know $x = [a, b]$ for some $a, b \in L$, so if we do some shenanigans:

$$\mathrm{ad}_{[a,b]} = [\mathrm{ad}_a, \mathrm{ad}_b]$$

Thus $\mathrm{Tr}(\mathrm{ad}_x) = 0$ and we know $\mathrm{ad}_x$ has a different eigenvector with eigenvalue $-\lambda$ called $z$.

By reality, either $\lambda$ is pure real or pure imaginary.

- Suppose $\lambda$ is pure real. Then, we can sacle such that $\lambda = 1$; the prior analysis goes through and we end up with

$$[x, y] = y, \qquad [z, x] = z, \qquad [y, z] = x.$$

- Otherwise, suppose $\lambda$ is pure imaginary. Then, we can scale such that $\lambda = i$. By basis change, we can make $\mathrm{ad}_x$ of the form

$$\mathrm{ad}_z = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

  i.e. $[x, y] = z$ and $[z, x] = y$. Now, we apply the Jacobi identity to get

$$[x, [y, z]] + [y, [z, x]] + [z, [x, y]] = 0 \implies [x, [y, z]] = 0$$

  Well, the only thing annihilated by $\mathrm{ad}_x$ is $x$ itself, so $[y, z] \in \mathbb{R}x$. By identical reasoning to above, I know that the trace of $\mathrm{ad}_y$ is 0, thus $[y, z] = x$ exactly. Then, we have

$$[x, y] = z, \qquad [z, x] = y, \qquad [y, z] = x.$$

# Solution 2 Automorphisms and derivations

**Problem 2.1** — Let $G$ be a Lie group with Lie algebra $\mathfrak{g}$. Let

$$\mathrm{Aut}(\mathfrak{g}) := \{g \in \mathbb{GL}(\mathfrak{g}) \mid [ga, gb] = g[a, b] \text{ for all } a, b \in \mathfrak{g}\}$$

be the algebra of **automorphisms** of $\mathfrak{g}$. Let

$$\mathrm{Der}(\mathfrak{g}) := \{g \in \mathfrak{gl}(\mathfrak{g}) \mid [x.a, b] + [a, x.b] = x.[a, b] \text{ for all } a, b \in \mathfrak{g}\}$$

be the Lie algebra of derivations of $\mathfrak{g}$. Prove that $\mathrm{Der}(\mathfrak{g})$ is the Lie algebra of $\mathrm{Aut}(\mathfrak{g})$.

Observe $\mathrm{Aut}(g)$ is a matrix subgroup. Recall that exp is a local homomorphism between a neighborhood of $0 \in \mathfrak{gl}(\mathfrak{g})$ and an open neighborhood of $1 \in \mathbb{GL}(\mathfrak{g})$.

Thus, sufficiently small matrices $M \in \mathfrak{gl}(\mathfrak{g})$ iff $\exp(M) \in \mathbb{GL}(\mathfrak{g})$, and we can expand this to second order:

$$\begin{aligned}[(I + tM + O(t^2))x, (I + tM + O(t^2))y] &= (I + tM + O(t^2))[x, y] \\ \implies t\left([Mx, y] + [x, My]\right) + O(t^2) &= tM[x, y]\end{aligned}$$

Taking the derivative at $t = 0$ gives $[Mx, y] + [x, My] = M[x, y]$, so the space of possible derivatives $f'(0)$ is exactly Der, as desired.

**Problem 2.2** — Show that $g \mapsto \mathrm{Ad}_g$ gives a morphism of Lie groups $G \to \mathrm{Aut}(\mathfrak{g})$.

Recall that any morphism of Lie groups induces a homomorphism of Lie algebras via the differential. In this case, consider the homomorphism $h \to ghg^{-1}$. Its differential is defined to be $\mathrm{Ad}_g$. So of course this is in $\mathrm{Aut}(\mathfrak{g})$, which is precisely the Lie homomorphisms $\mathfrak{g} \to \mathfrak{g}$.

**Problem 2.3** — Show that $x \mapsto \mathrm{ad}_x$ is a morphism of Lie algebras $\mathfrak{g} \to \mathrm{Der}(\mathfrak{g})$.

I mean, we're trying to show

$$[\mathrm{ad}_x\, a, b] + [a, \mathrm{ad}_x\, b] = \mathrm{ad}_x[a, b]$$

but we know $\mathrm{ad}_x = [x, \bullet]$ so this is just the Jacobi identity:

$$[[x, a], b] + [a, [x, b]] = [x, [a, b]]$$

**Problem 2.4** — Show that $\mathrm{ad}(\mathfrak{g})$ is an ideal in $\mathrm{Der}(\mathfrak{g})$.

For any $M \in \mathrm{Der}$, $\mathrm{ad}_x \in \mathrm{ad}(\mathfrak{g})$, and $y \in \mathfrak{g}$, we have

$$M[x, y] - [x, My] = [Mx, y]$$

because $M$ is a derivation. Thus $[\mathrm{ad}_x, M] = Mx \in \mathrm{ad}(\mathfrak{g})$, as desired.

# Solution 3 Real forms

**Problem 3.1 —** Let $G$ be a complex connected simply connected Lie group, with Lie algebra $\mathfrak{g}$, and let $\mathfrak{k} \subset \mathfrak{g}$ be a real form of $\mathfrak{g}$.

Define the $\mathbb{R}$-linear map $\theta : \mathfrak{g} \to \mathfrak{g}$ by $\theta(x + iy) = x - iy$, with $x, y \in \mathfrak{k}$. Show that $\theta$ is an automorphism of $\mathfrak{g}$ (considered as a real Lie algebra), and that it can be uniquely lifted to an automorphism $\theta : G \to G$ of the group $G$ (considered as a real Lie group).

$\theta : x + iy \mapsto x - iy$ is clearly $\mathbb{R}$-linear, and

$$\begin{aligned}\theta([a + bi, x + yi]) &= \theta([a, x]) - \theta([b, y]) + \theta(i[b, x]) + \theta(i[a, y]) \\ &= [a, x] - [b, y] - i[b, x] - i[a, y] = [a - bi, x - yi] \\ &= [\theta(a + bi), \theta(c + di)]\end{aligned}$$

thus $\theta$ is a $\mathbb{R}$-homomorphism of Lie algebras. (We use the fact that $\theta$ fixes $\mathfrak{k}$ and negates $i\mathfrak{k}$).

Since $G$ is simply connected, Lie's second theorem tells us that every automorphism of its Lie algebra corresponds bijectively to a unique lifting to an automorphism $\Theta : G \mapsto G$. In particular, $\Theta$ should send $e^{t(a+bi)}$ to $e^{t\theta(a+bi)}$.

Since $\theta$ was a $\mathbb{R}$-automorphism, $\Theta$ is an automorphism of $G$ as a real Lie group.

**Problem 3.2 —** Let $K = G^\theta$. Show that $K$ is a real Lie group with Lie algebra $k$.

Since $\Theta$ is an automorphism, the set of fixed points $K = G^\Theta$ is a subgroup. $\Theta$ is continuous, thus its fixed points form a closed set, hence $K$ is a closed (real) Lie subgroup.

$K$ is fixed by $\Theta$, thus $\mathrm{Lie}(K)$ is fixed by $\theta$; the differential of the map $K \to K$ has to map into $\mathrm{Lie}(K)$. Thus, $\mathrm{Lie}(K) \subseteq \mathfrak{k}$.

On the other hand, for any $x \in \mathfrak{k}$, $e^{tx}$ gets sent to $e^{t\theta(x)} = e^{tx}$ by $\Theta$, so this path is in $K$. Thus $\mathfrak{k} \subseteq \mathrm{Lie}(K)$, and we're done.

# Solution 4 Decomposing $\mathfrak{sl}_3(\mathbb{R})$ into $\mathfrak{sl}_2(\mathbb{R})$ submodules

**Problem 4.1 —** Let $M = \mathfrak{sl}_3(\mathbb{R})$ contains a copy of $\mathfrak{sl}_2$ in its upper left had $2 \times 2$ position. Write $M$ as direct sum of irreducible $\mathfrak{sl}_2$-submodules ($M$ viewed as $\mathfrak{sl}_2$ module via the adjoint representation).

Here you go:

$$\left\{\begin{pmatrix} a & & \\ & a & \\ & & -2a \end{pmatrix}\right\} \oplus \left\{\begin{pmatrix} a & b & \\ c & -a & \\ & & 0 \end{pmatrix}\right\} \oplus \left\{\begin{pmatrix} 0 & 0 & a \\ 0 & 0 & b \\ 0 & 0 & 0 \end{pmatrix}\right\} \oplus \left\{\begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ a & b & 0 \end{pmatrix}\right\}$$

These subspaces are clearly linearly independent. They also span $\mathfrak{sl}_3$ because of dimension counting: $1 + 3 + 2 + 2 = 8$. Now we do commutation with copies of $\mathfrak{sl}_2$ in the upper left hand corner to show that these are invariant and irreducible.

1. The first subspace commutes with all of $\mathfrak{sl}_2$ and is thus invariant; it is also one-dimensional and thus irreducible.

2. The second subspace is $\mathfrak{sl}_2$; recall that $[\mathfrak{sl}_2, \mathfrak{sl}_2] = \mathfrak{sl}_2$ so this is irreducible.

3. The second subspace is acted upon as

$$\begin{pmatrix} M & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & 0 & a \\ 0 & 0 & b \\ 0 & 0 & 0 \end{pmatrix} - \begin{pmatrix} 0 & 0 & a \\ 0 & 0 & b \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} M & 0 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 & a' \\ 0 & 0 & b' \\ 0 & 0 & 0 \end{pmatrix}$$

   (Here $M \in \mathfrak{sl}_2$ and $M\binom{a}{b} = \binom{a'}{b'}$). There are no eigenvectors of *all of* $\mathfrak{sl}_2$, so this is irreducible.

4. The fourth subspace is the same as the third.

# Solution 5 Solvable and nilpotent Lie algebras

**Problem 5.1** — Let $L$ be a Lie algebra. Prove that $L$ is solvable if and only if $\operatorname{ad} L$ is solvable.

Recall that for any two Lie subalgebras of $L$, $K$ and $H$, $\operatorname{ad}[K, H] = [\operatorname{ad} K, \operatorname{ad} H]$. Thus, we can just pull out of the entire derived series:

$$[[\operatorname{ad} L, \operatorname{ad} L], [\operatorname{ad} L, \operatorname{ad} L]] = [\operatorname{ad}[L, L], \operatorname{ad}[L, L]] = \operatorname{ad}[[L, L], [L, L]]$$

Now, if $\operatorname{ad} L$ is solvable, then $0 = (\operatorname{ad} L)^{(k)} = \operatorname{ad} L^{(k)}$. This literally says $[L^{(k)}, L] = 0$, thus $L^{(k+1)} = 0$.

If $L^{(k)} = 0$, then $(\operatorname{ad} L)^{(k)} = 0$ automatically.

**Problem 5.2** — Prove that $L$ is nilpotent if and only if $\operatorname{ad} L$ is nilpotent.

Similarly, $\operatorname{ad} L^k = (\operatorname{ad} L)^k$. If $(\operatorname{ad} L)^k = 0$, then $\operatorname{ad} L^k = 0$, thus $[L, L^k] = 0$, which means $L^{k+1} = 0$ as desired. If $L^k = 0$, then $(\operatorname{ad} L)^k = 0$ done.

**Problem 5.3** — Prove that $L$ is solvable if and only if there exists a chain of subalgebras

$$L = L_0 \supset L_1 \supset L_2 \supset \cdots \supset L_k = 0$$

such that $L_{i+1}$ is an ideal of $L_i$ and such that each $L_i/L_{i+1}$ is abelian.

If $L$ is solvable, then let $L_k = L^{(k)}$; in general $L/[L, L]$ is abelian thus we're done.

Now, if $K \subset L$ is an ideal such that $L/K$ is abelian, then $[L, L] \subset K$. Indeed, for all $x, y \in L$, $0 = [x + K, y + K] = [x, y] + K$ thus $[x, y] \in K$.

By induction, if such a chain exists, $L^{(k)} \subset L_k = 0$, so we're done.